Question 5: Initialize a repository, stage and commit changes, create branches, merge branches, and push the repository to GitHub.

Marks: 5 | Course outcome: CO1 | Environment: Ubuntu 24.04 LTS (WSL2), Git 2.43.0, Jenkins 2.580.2 LTS

The complete version-control workflow was performed on the sample Flask application repository and pushed to GitHub (https://github.com/meenakshi16rp/Devops-lab-1-).

1. Initialize, stage and commit

init, stage, commit

```
cd ~/ensp461-lab1
git init -b main
git add README.md .gitignore app/app.py app/requirements.txt app/tests/test_app.py
git commit -m "Initial commit: sample Flask application with unit tests"
git add deploy/deploy.sh
git commit -m "Add automated deployment script with health verification"
```

2. Branch, feature work and merge

branch and merge (--no-ff)

```
git checkout -b feature/app-info-endpoint
# implement /info endpoint + unit test on the branch
git add app/ && git commit -m "Add /info endpoint with application metadata and unit test"
git checkout main
git merge --no-ff feature/app-info-endpoint -m "Merge feature/app-info-endpoint: add /info \
      endpoint to main"
```

commit history after the merge (git log --graph)

```
=== git log --oneline --graph --all ===
* eaa01d3 Keep deployed app running: set JENKINS_NODE_COOKIE to bypass process tree killer
* c9df47c Fix deploy script: reuse pipeline virtualenv and add health-check retry loop
* eedda81 Add declarative Jenkinsfile with build, test and deploy stages
*   b4ead4e Merge feature/app-info-endpoint: add /info endpoint to main
|\
| * 946c34c Add /info endpoint with application metadata and unit test
|/
* 44c816b Add automated deployment script with health verification
* 6b3bf69 Initial commit: sample Flask application with unit tests

=== git config --global --list ===
user.name=Meenakshi
user.email=meenakshi16rp@gmail.com
init.defaultbranch=main
core.editor=nano
credential.helper=store

=== git remote -v ===
origin□https://github.com/meenakshi16rp/Devops-lab-1-.git (fetch)
origin□https://github.com/meenakshi16rp/Devops-lab-1-.git (push)

=== git branch -a ===
  feature/app-info-endpoint
* main
  remotes/origin/feature/app-info-endpoint
  remotes/origin/main
```

3. Push to GitHub

remote and push

```
git remote add origin https://github.com/meenakshi16rp/Devops-lab-1-.git
git push -u origin main
git push -u origin feature/app-info-endpoint
```

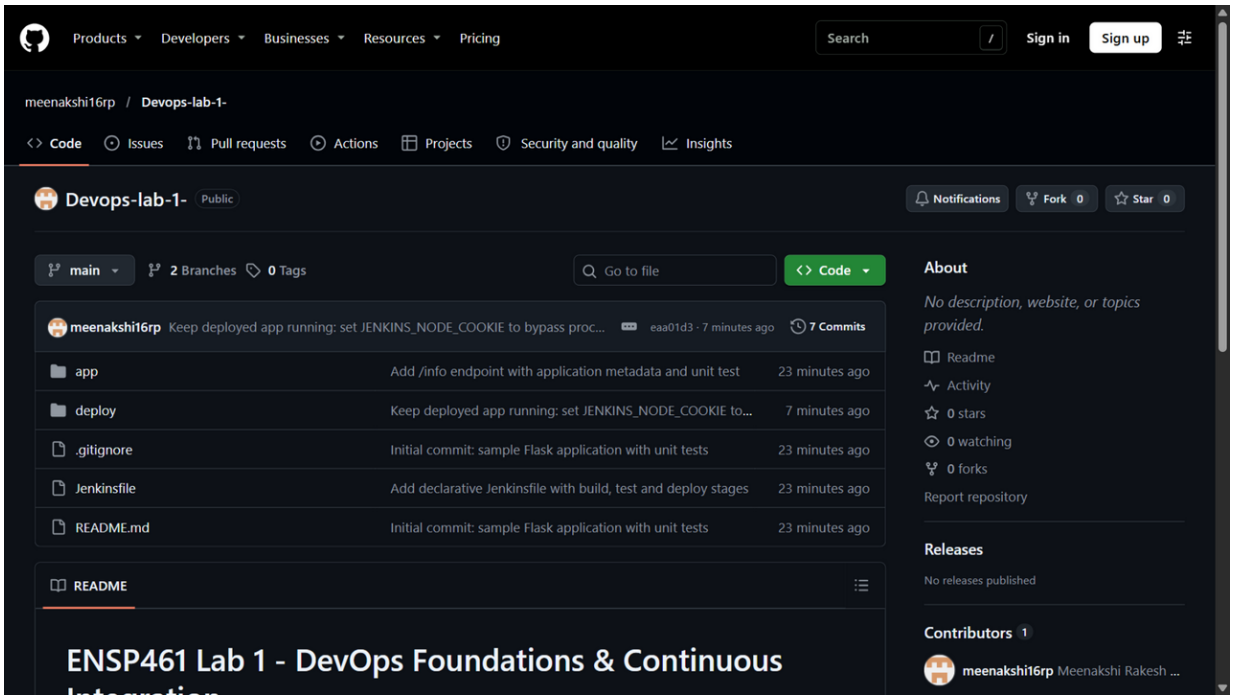


Figure: Repository on GitHub: 7 commits, 2 branches, Jenkinsfile and README visible

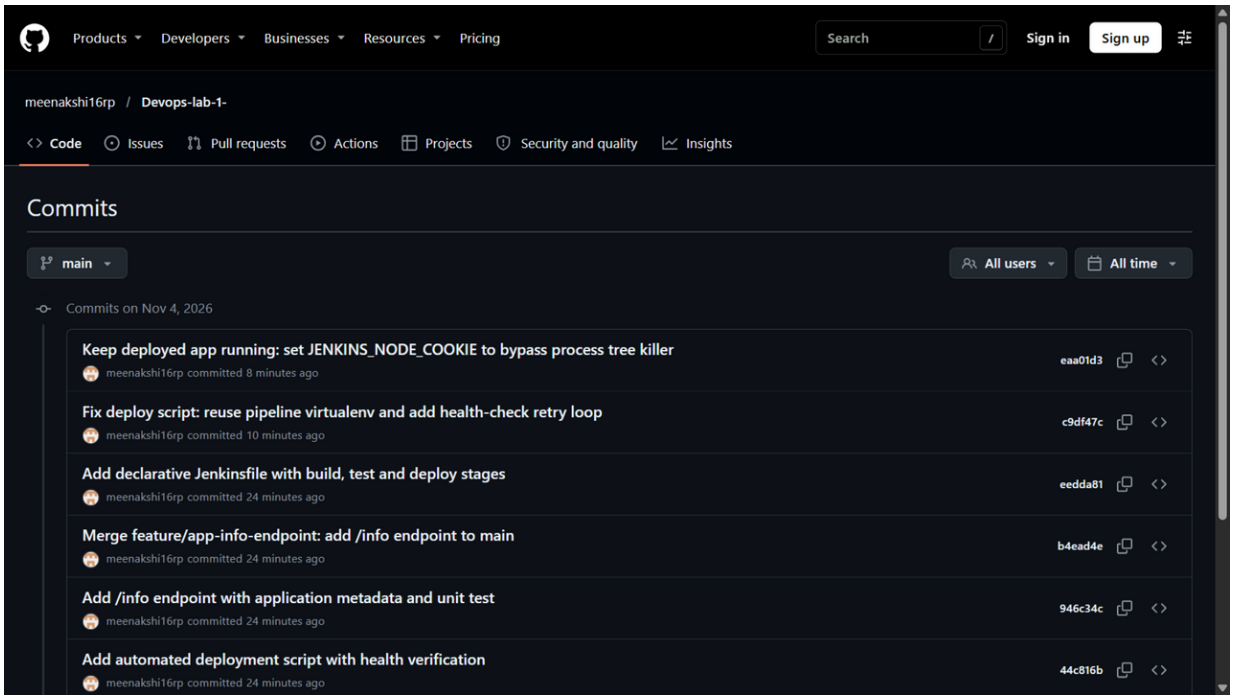


Figure: Commit history on GitHub - merge commit and feature branch visible

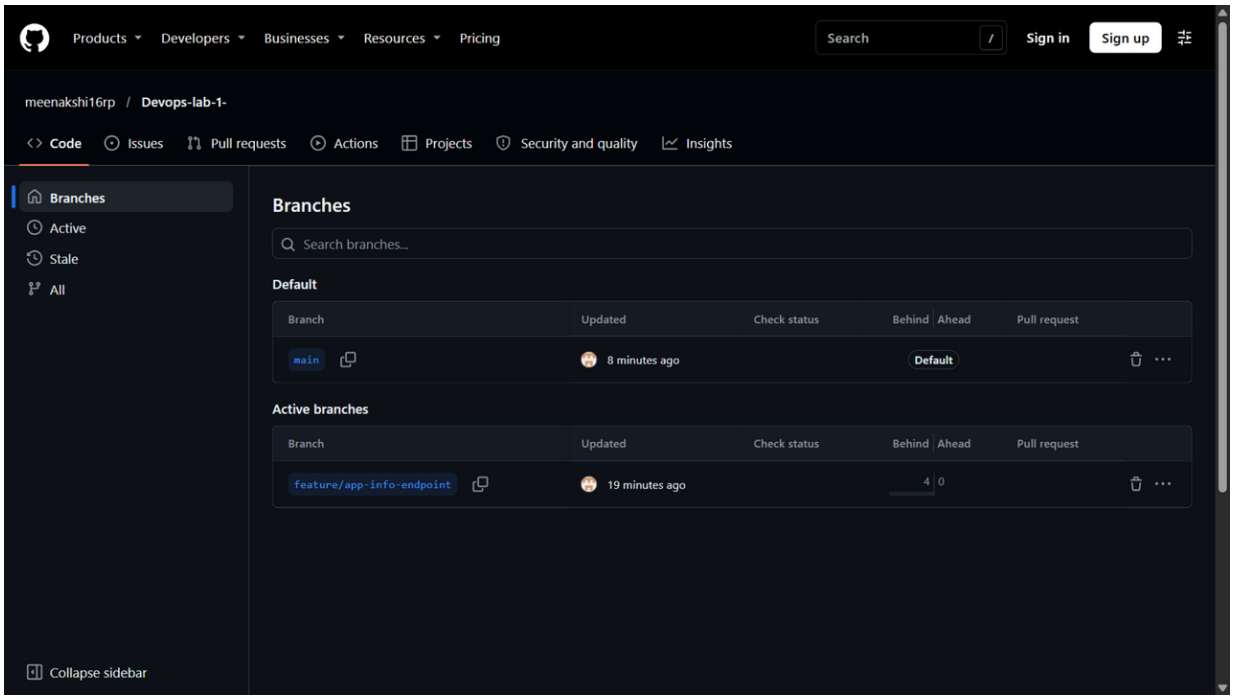


Figure: Branches page: main (default) and feature/app-info-endpoint

4. Observations

- --no-ff merging keeps the feature branch visible in history, which documents the workflow.
- Every commit message describes what changed so the log reads as a narrative.
- The feature branch remains on GitHub as evidence of the branching workflow.
- credential.helper store keeps the PAT outside the repository (in ~/.git-credentials, chmod 600).